Chapter 4: Quantitative Approaches in Research

Statistics, Population, Sampling

What is statistics in research?

In social science based research, statistics is the scientific method for **collecting, organizing, analyzing, interpreting, and presenting data** to extract meaningful information, draw conclusions, and make informed decisions about a larger population based on a sample.

It provides a framework and tools for understanding **patterns, relationships, and uncertainties** within data, allowing researchers to communicate findings with credibility and contribute to scientific knowledge.

Sampling definitions

Population: The total number of individuals or things you are interested in for your research

Sample: A smaller number within the population that will represent the whole population

Sampling: The process and method of selecting your sample

Probability vs Non-probability

Probability sampling (also known as random sampling) is a kind of sample selection where randomization is used instead of deliberate choice. Each member of the population has a known, non-zero chance of being selected.

Non-probability sampling techniques are where the researcher deliberately picks items or individuals for the sample based on non-random factors such as convenience, geographic availability, or costs.

Simple Random Sampling

In simple random sampling, every element in the population has an **equal chance** of being selected as part of the sample. Simple random sampling can be done by anonymizing the population – e.g. by assigning each item or person in the population a number and then picking numbers at random.

Pros: Simple random sampling is easy to do and cheap. Designed to ensure that every member of the population has an equal chance of being selected, it reduces the **risk of bias** compared to non-random sampling.

Cons: It offers no control for the researcher and may lead to unrepresentative groupings being picked by chance.

Systematic Sampling

With systematic sampling, the random selection only applies to the first item chosen. A rule then applies so that every nth item or person after that is picked.

PROCESS: Sort your list in a random way to ensure that selections won't be accidentally clustered together OR order your list alphabetically by first name and then pick every fifth name to eliminate bias.

Next, you need to decide your sampling interval – for example, if your sample will be 10% of your full list, your sampling interval is one in 10 – and pick a random start between one and 10 – for example three. This means you would start with person number three on your list and pick every tenth person.

Pros: Systematic sampling is efficient and straightforward, especially when dealing with populations that have a clear order. It ensures a uniform selection across the population.

Cons: There's a potential risk of introducing bias if there's an unrecognized pattern in the population that aligns with the sampling interval.

Stratified Sampling

Stratified sampling consists of random selection within predefined categories or groups.

PROCESS: Stratification by gender for example

Pros: Stratified sampling enhances the representation of all identified subgroups within a population, leading to more accurate results in heterogeneous populations.

Cons: This method requires accurate knowledge about the population’s stratification, and its design and execution can be more intricate than other methods.

Cluster Sampling

With cluster sampling, groups rather than individual units of the target population are selected at random for the sample. These might be pre-existing groups, such as people in certain regions or students belonging to an academic session.

Cluster sampling can be done by selecting the entire cluster, or in the case of two-stage cluster sampling, by randomly selecting the cluster itself, then selecting at random again within the cluster.

Pros: Cluster sampling is economically beneficial and logistically easier when dealing with vast and geographically dispersed populations.

Cons: Due to potential similarities within clusters, this method can introduce a greater sampling error compared to other methods.

Non-Probability Sampling

The non-probability sampling methodology doesn't offer the same bias-removal benefits as probability sampling, but there are times when these types of sampling are chosen for expediency or simplicity.

Convenience sampling

People or elements in a sample are selected on the basis of their accessibility and availability. This kind of sample can have value, especially if it's done as an early or preliminary step, but significant bias will be introduced.

Pros: Convenience sampling is the most straightforward method, requiring minimal planning, making it quick to implement.

Cons: Due to its non-random nature, the method is highly susceptible to biases, and the results are often lacking in their application to the real world.

Purposive sampling

This process selects a sample based on pre-existing knowledge of the population, its elements, and the nature of our research aims

Researchers depend on their understanding of the population, selecting based on the criteria relevant to their research enquiries

Risk of researcher bias in selection of sample is eminent

Snowball sampling

Used mainly in field observation studies or qualitative interviewing

The snowball sampling method begins by identifying a single subject or small number of subjects and then asking the subject/subjects to identify others

Helpful in scenarios where the researchers are outsiders to the community

Quota Sampling

Like the probability-based stratified sampling method, this approach focuses on who should be recruited for a survey according to certain groups or criteria.

For example, the quota might include a certain number of males and a certain number of females. Alternatively, you might want your samples to be at a specific income level or in certain age brackets or ethnic groups.

Pros: Quota sampling ensures certain subgroups are adequately represented, making it great for when random sampling isn't feasible but representation is necessary.

Cons: The selection within each quota is non-random and researchers' discretion can influence the representation, which both strongly increase the risk of bias.